

ICT Health and Safety Program

Part I. All ICT Users

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The ICT Formula



People;
Leading Faculty
Motivated Students
Experienced Engineers

Process;
Safe and cutting-edge
Unparalleled facilities

Product;
Implementable
Comprehensive
Game-changing

Responsibilities and Expectation

- **Faculty, Investigators and Research Engineers are responsible for;**
 - Maintaining a healthy and safe environment
 - Providing appropriate training
 - Monitoring lab users to address safety
- **Students and Visitors are responsible for;**
 - Complying with ALL applicable health and safety requirements
 - Protecting their own health and safety and that of the other ICT users
 - Following test procedures and operating equipment appropriately

Minimum Laboratory Safety Training



Risk Assessment

- **Use the following approach**
 - **R** – recognize the hazards
 - **A** – assess the risks
 - **M** – mitigate the risks
 - **P** – prepare for emergencies
 - Know nearest eyewash and shower stations
 - Nearest fire extinguishers and exits

Risk Assessment

■ Lab Risk Assessments



○ <https://www.youtube.com/watch?v=TCHCxE7ygXo&t=8s>

A. Recognize the Hazards

■ Hazard:

- An intrinsic property of a substance/condition that can result in potential harm/damage to an individual or thing
- It cannot be changed

■ Risk

- The probability that exposure to a hazard will result in adverse consequences

Risk level = probability of exposure to hazard x severity of hazard

B. Assess the Risks

- **Failure Mode and Effects Analysis (FMEA)**
 - Tool assigns **numbers** to **probability** and **severity** to calculate where the probability falls on the risk matrix
 - **Analysis** is performed for **each step** of a procedure to identify those that pose a high or unacceptable risk
 - **Further evaluation** is required for;
 - Failure that results in **critical** or **catastrophic** consequences, or
 - Any **score** above **63**

B. Assess the Risks – Example

- E.g What is the risk of riding a bicycle in a dark street with no lights on

Probability (P)
1 = Extremely unlikely
3 = Remote
5 = Occasional failure
7 = Repeated
9 = Frequent failure

Severity (S)
1 = No effect
2 = Very minor effect
3 = Light injuries
16 = Serious injuries
25 = Critical
49 = Catastrophic

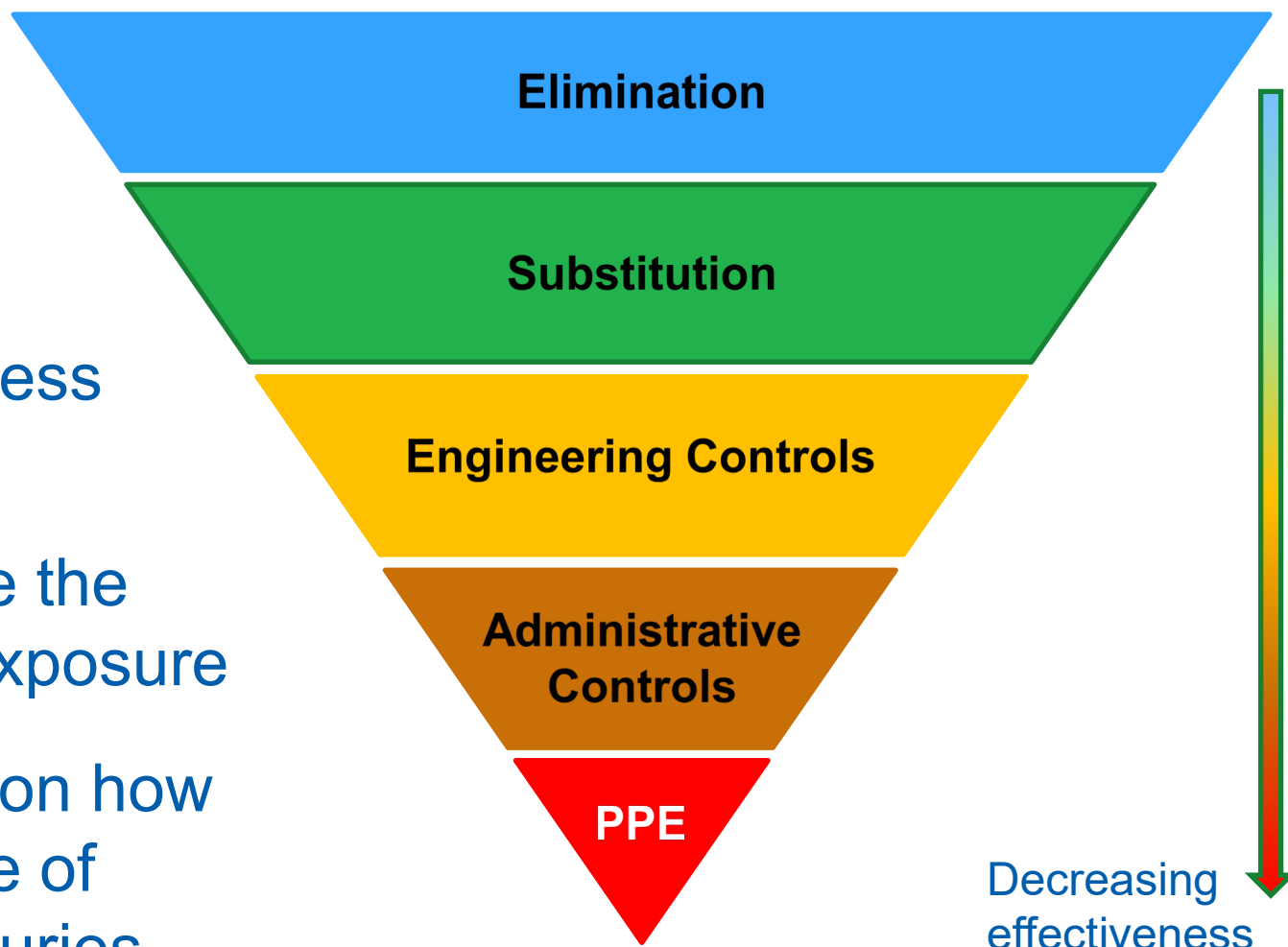
Consequences
Acceptable ≤ 27
$28 \leq \text{Moderate} \leq 63$
$75 \leq \text{High} \leq 125$; High = 49 when S = 49
Unacceptable ≥ 144

- Failure event: Getting hit by a car
- $P = 5$; $S = 49$; Therefore risk level ($P \times S$) = 245 unacceptable

C. Minimize the Risks

■ Hierarchy of Controls

- Elimination can be difficult to implement
- Substitution uses something less hazardous
- Engineering controls separate the user from hazard or reduce exposure
- Administrative controls focus on how people work to reduce chance of human error, failures , and injuries



C. Minimize the Risks – Example

- After completing FMEA, it is important to determine how to **reduce the risk**. Risk can be reduced in two ways;
 - Lower the probability of getting hit e.g by having lights
 - P changes from 5 to 1, but **S = 49**, **risk level = 49** unacceptable
 - Or **reduce the severity** e.g by wearing a helmet
 - P = 5, but S changes from 49 to 16, **risk level = 64** high
 - Implementing both, P = 1, S = 16, risk level = 16 Acceptable

C. Minimize the Risks – Administrative Controls



C. Minimize the Risks – Administrative Controls

- **NEVER** perform work you are not comfortable with
- **Housekeeping is a MUST**
 - All samples must have labels (project, student name, dates)
 - Workbenches, countertops, and carts are for work not storage
 - Return items/tools to appropriate location e.g solvents to flammables cabinet
 - Counters and shelves must be lined with paper

C. Minimize the Risks – Administrative Controls

- If you open a door/window, make sure you close that door/window
- Counters and shelves **MUST** be lined with paper
- **The last TWO people to leave must set the alarm**
- **If you think it is likely not the best idea, DON'T DO IT!**

C. Minimize the Risks – PPE



C. Minimize the Risks – PPE

- Wear **EYE PROTECTION**,
- **SAFETY TOE SHOES, STURDY PANTS** at all times when in;
 - Main Lab, MPF, and ATLAS Yard
 - Safety glasses = ANSI Z87+; Shoes = ANSI Z41.1
 - Sturdy clothing: No sweat pants, jogging pants, pajamas, yoga pants. No shorts of any kind
 - No loose fitting or low cut shirts
 - Long hair must be tied back. No loose or dangling jewelry



C. Minimize the Risks – PPE

- **Safety glasses test**



- <https://www.youtube.com/watch?v=0TOymIHZDb8>

C. Minimize the Risks – PPE

■ Respiratory Protection

- Before you use a respirator (except paper dust mask), the following steps must be taken:
 - Contact UIUC Occupational Health and Safety to schedule evaluation of work area
 - Complete a medical evaluation
 - Complete respiratory protection training



D. Emergency Preparedness – Contacts

- Emergency Contact

On ANY red phone
Call 911



D. Emergency Preparedness – Contacts

■ Non- Emergency Contact



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D. Emergency Preparedness – Introduction

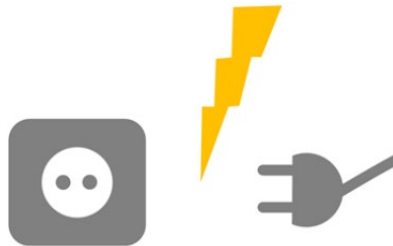
■ Know the following

- Nearest exits
- Nearest first aid kits
- Nearest eyewash and safety shower stations
- Location of fire extinguishers and fire alarm pull stations
- Tornado shelter



D. Emergency Preparedness – Fire

■ Causes of fires

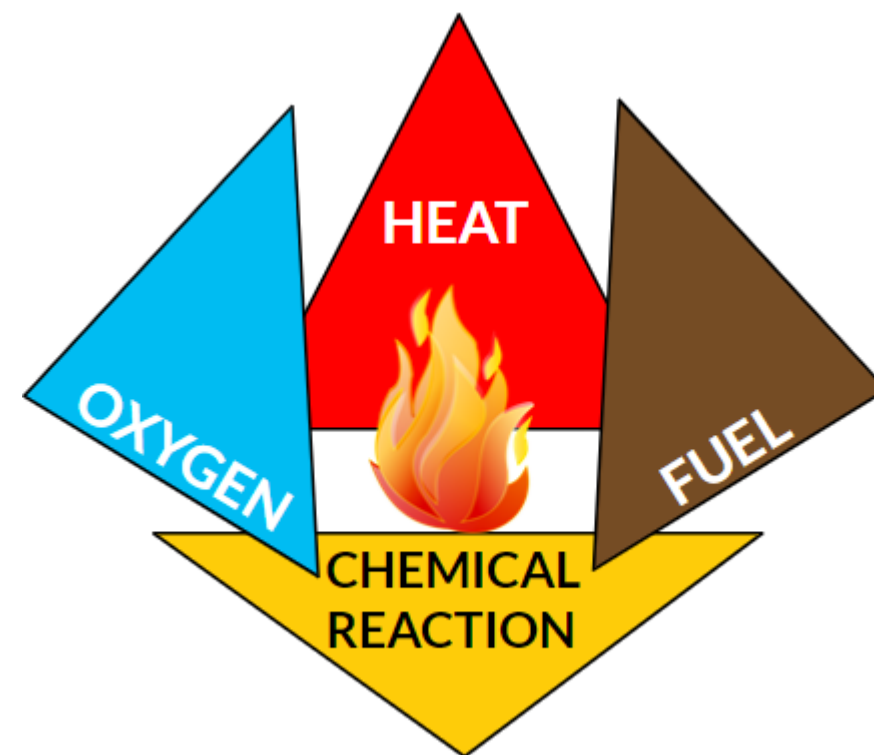
-  Auto-ignition of combustible materials
- Ignition of flammable materials 
- Smoldering materials improperly disposed
- Overheating of bad electrical connections 



D. Emergency Preparedness – Fire

■ Fire tetrahedron – components of a fire

- Fuel – source for the combustion
- Oxygen – sustains the combustion
- Chemical reaction
 - Between fuel and oxygen
 - Results in heat
- Firefighting removes one element



D. Emergency Preparedness – Fire

■ Classes of fires

- Class **A** fires – paper, wood, trash, plastic



- Class **B** fires  – flammable liquids, gasoline, paint, flammable gases

- Class **C** fires  – caused by energized electrical equipment

D. Emergency Preparedness – Fire

■ In case of a fire

- Pull the fire alarm
- Call 911
- Make a quick decision
 - Whether to attempt to extinguish the fire
 - Or to evacuate
 - If in **DOUBT**, Get **OUT!!**



D. Emergency Preparedness – Fire

- **Only fight a fire if**
 - Fire is small and contained
 - You have an escape route
 - You are safe from toxic smoke
 - You are up to date on training
 - Your instincts tell you it's ok



D. Emergency Preparedness – Fire

■ Use of fire extinguishers

- ICT has dry chemical (ABC) fire extinguishers
 - Used to extinguish classes A, B, or C fires
- **P** – Pull the pin
- **A** – Aim at the base of the fire
- **S** – Squeeze the trigger to discharge
- **S** – Sweep the nozzle from side to side

■ **Remember: Stay 8 to 10 ft from the fire**



D. Emergency Preparedness – Fire

- **P. A. S. S**



- <https://www.youtube.com/watch?v=aU35ioqiJmA>

D. Emergency Preparedness – Tornado

- **What you need to know**
 - Tornado Conditions
 - Designated Tornado Shelters
 - What to do once inside the Shelter
 - What to do if caught outside



D. Emergency Preparedness – Tornado

■ Tornado conditions

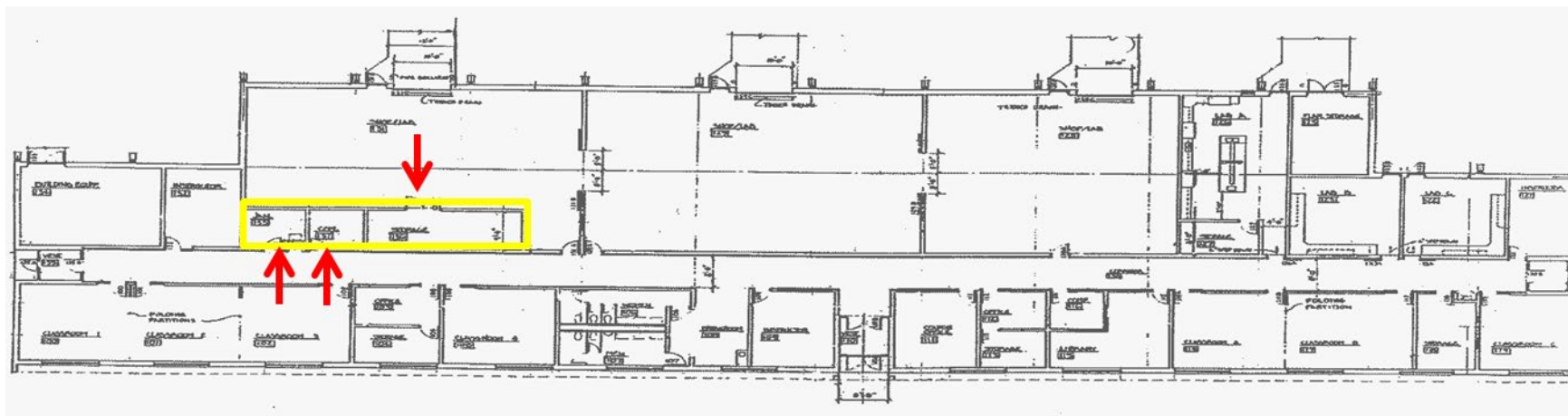
- Tornado Watch: Be Alert
 - Conditions are favorable for the development of severe thunderstorms and tornadoes
- Tornado Warning: Seek Shelter
 - Strong rotation in a thunderstorm is indicated by Doppler weather radar
 - Or a tornado has been sighted by spotters



D. Emergency Preparedness – Tornado

■ Designated Tornado Shelters

- Main building
 - Janitorial closet (across from student offices)
 - Highbay A tool room



- MPF: Restroom

D. Emergency Preparedness – Tornado

■ Once inside the Shelter

- DUCK: Assume a seated position with head down and hands locked over your head
- AND COVER: Cover your upper body with coat or heavy material for protection from flying debris
 - Stay close to the floor
 - Stay away from outside walls, doors, windows, and highbay areas
- If caught outside, find shelter in a ditch or remain in your vehicle and cover your head for protection



D. Emergency Preparedness – First Aid

- **Emergency call 911**
- **Inform Research Engineers if you use any items from the first aid kits so that they can be replenished**
- **Report any injury to Research Engineers immediately**



D. Emergency Preparedness – Electrical Safety

- Do not try to pull victim away from circuit
- Call for help !!
- De-energize the circuit or pry person away from circuit with non-conductive device
- If the person is not breathing start CPR
- Wait with victim until help arrives



D. Emergency Preparedness – Electrical Safety



D. Emergency Preparedness – First Aid

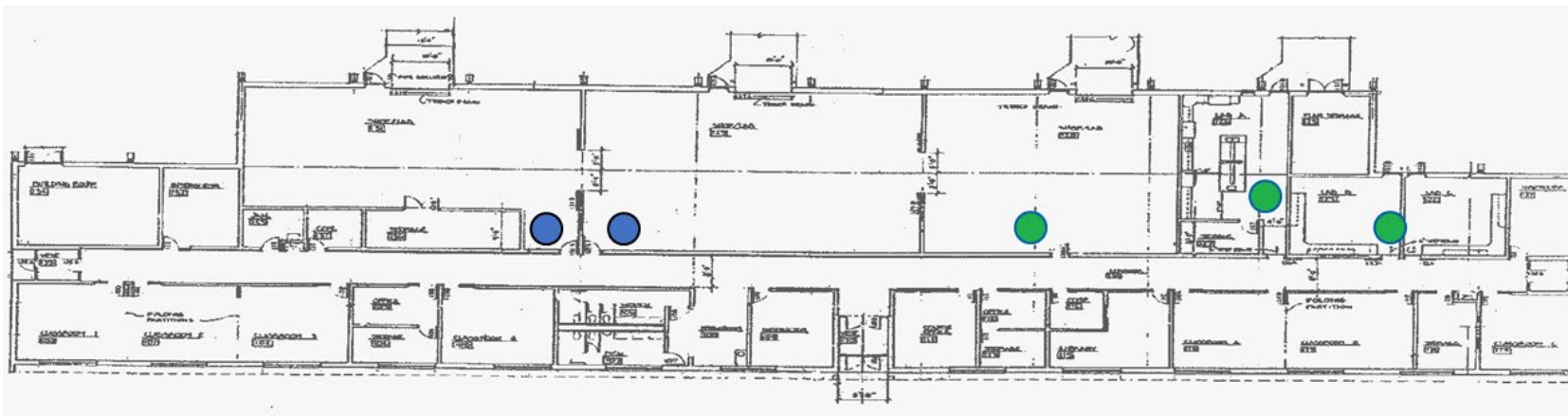
■ First Aid kit locations at ICT

- Lab A (binder) – East Wall
- Lab B (volumetrics) – South Door
- Highbay A – South Door
- Highbay B (near servoroom) – South Door
- Highbay C (mixing) – South Door



D. Emergency Preparedness – First Aid

■ First Aid kit locations at ICT



■ First Aid kit locations in MPF

- Bathroom
- Saw room

Other ICT Policies – ICT Use Requirements

- **Complete the following;**
 - DRS online “general lab safety” training and provide certificate of completion <http://research.illinois.edu/>
 - Review the UIUC Laboratory Safety Guide <https://www.drs.illinois.edu/site-documents/LaboratorySafetyGuide.pdf>
 - Read and understand ICT safety rules and complete the compliance form in Appendix D
 - Submit all completed forms to Research Engineers

Other ICT Policies – ICT Use Requirements

- Training is required to operate any equipment at ICT
- Request and schedule training with Research Engineers
- No Equipment is to be used without training by the Research Engineers or by a person designated by them
- Violations will result in **SUSPENSIONS** from the lab

Other ICT Policies – ICT Use Requirements

■ Tag out for Maintenance

- Indicates device (valve, switch, etc.) should not be altered

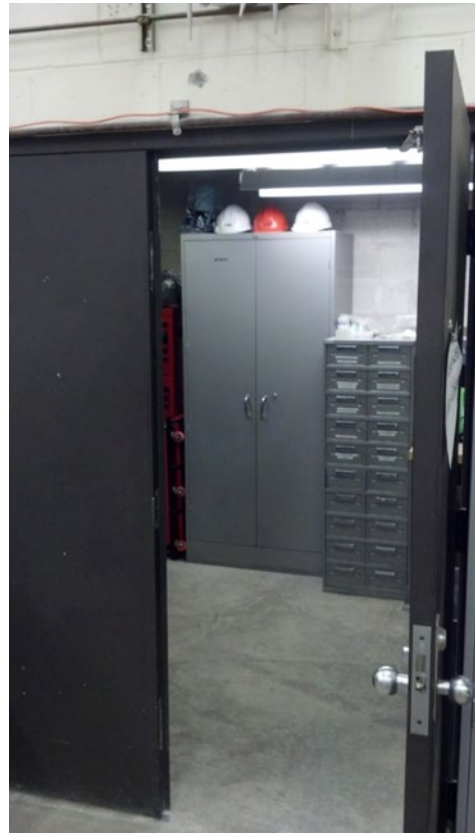


Other ICT Policies – ICT Access

- **After submitting safety documents;**
 - Provide Name, Net ID, UIN
 - You get swipe access to the main building from 8am to 5pm
 - Access to MPF is based on the individual's research role on a case-by-case basis

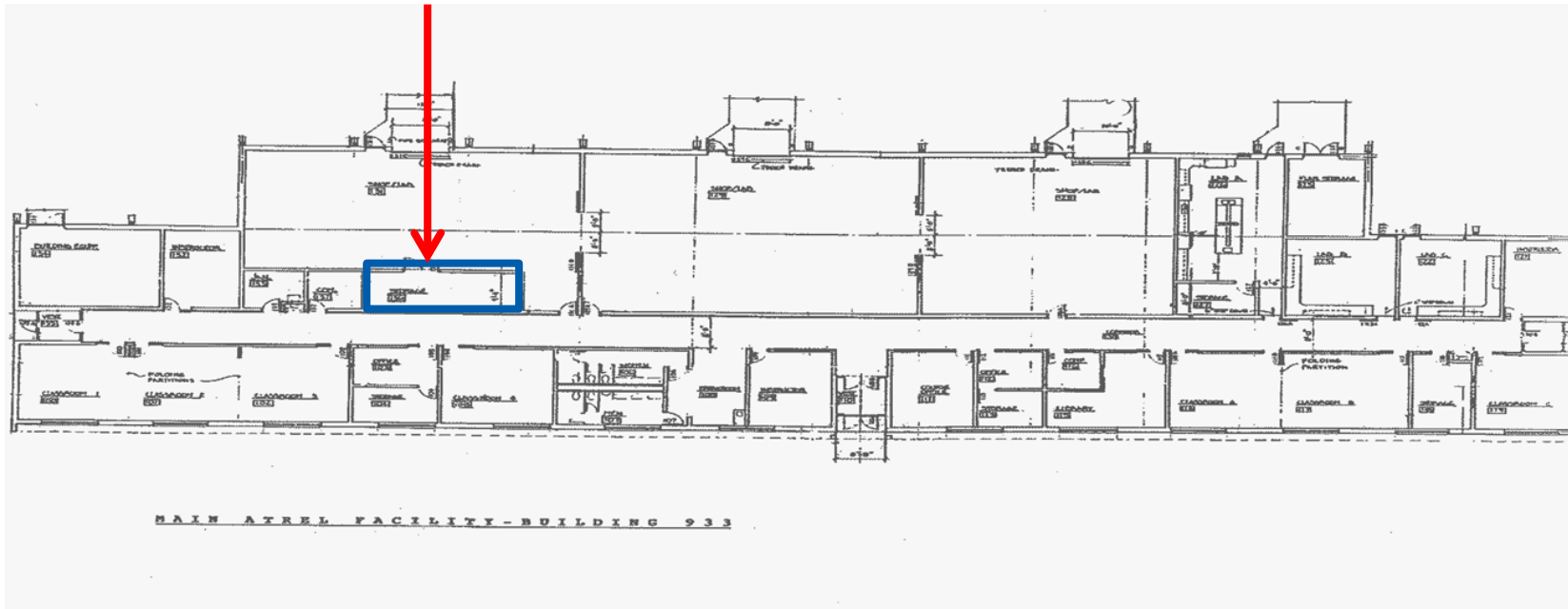
Other ICT Policies – General Supplies

- In the main building general supplies are in the tool room in Highbay A
- Let us know when an item is running low



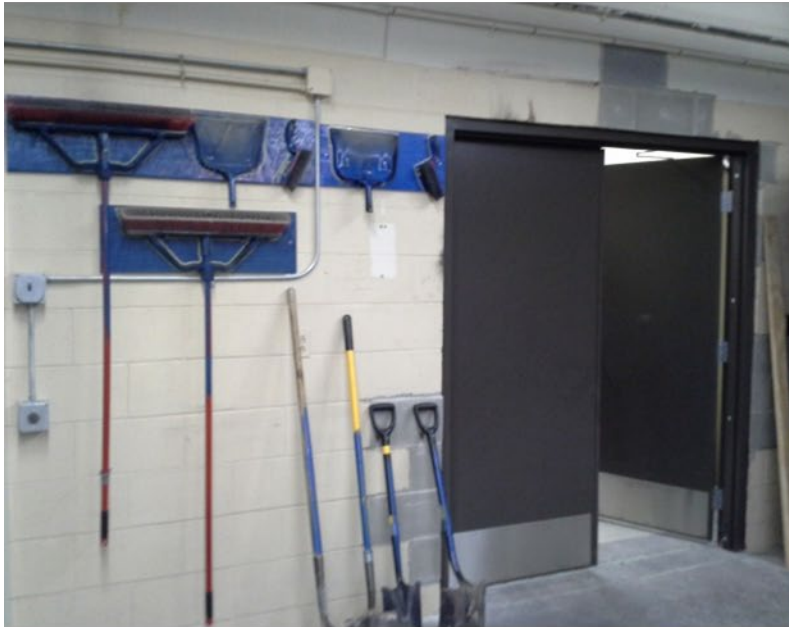
Other ICT Policies – General Supplies

■ Location of ICT Tool room



Other ICT Policies – MPF Tool room

- Keep cleaning items and tools organized in MPF



Remember

**“Don’t learn
Safety
by accident”**



THANK YOU

Any Questions?

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[Illinois Center for Transportation](#)

